

Granting Non-AI Experts Creative Control Over AI Systems

Michelle S. Lam
Stanford University
Stanford, CA, USA
mlam4@cs.stanford.edu

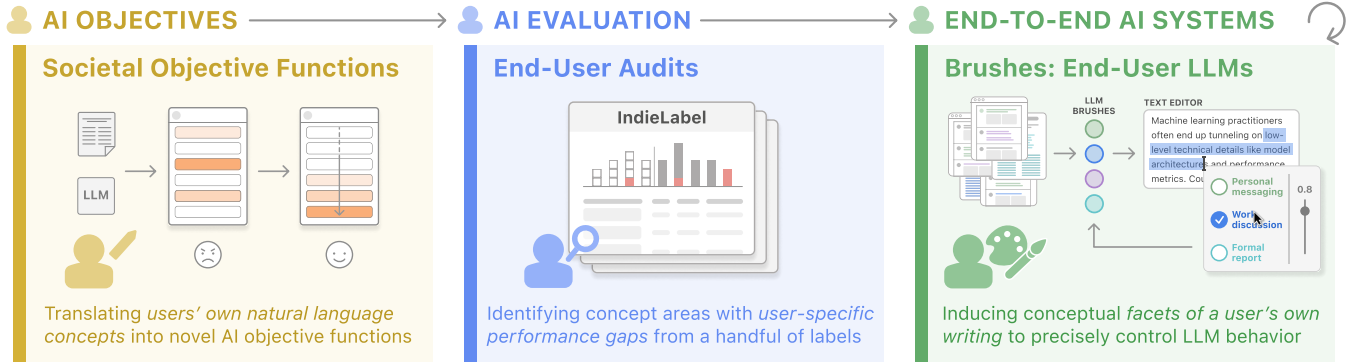


Figure 1: My research develops systems for users to creatively define AI behavior in terms of personalized, self-defined concepts. These tools give non-AI experts control over the full AI lifecycle: (1) designing AI *objective functions*, (2) leading AI *evaluations*, and (3) controlling AI systems *end-to-end* from modeling to application to evaluation and repair.

ABSTRACT

Many harmful behaviors and problematic deployments of AI stem from the fact that AI experts are *not* experts in the vast array of settings where AI is applied. Non-AI experts from these domains hold promising potential to contribute their expertise and directly design the AI systems that impact them, but they face substantial technical and effort barriers. Could we redesign AI development tools to match the language of non-technical end users? My research develops novel systems allowing non-AI experts to define AI behavior in terms of interpretable, self-defined concepts. Monolithic, black-box models do not yield such control, so we introduce techniques for users to create many narrow, personalized models that they can better understand and steer. We demonstrate the success of this approach across the AI lifecycle: from designing AI objectives to evaluating AI behavior to authoring end-to-end AI systems. When non-AI experts design AI from start to finish, they notice gaps and build solutions that AI experts could not—such as creating new feed ranking models to mitigate partisan animosity, surfacing underreported issues with content moderation models, and activating unique pockets of LLM behavior to amplify their personal writing style.

KEYWORDS

human-AI interaction, responsible AI, large language models

ACM Reference Format:

Michelle S. Lam. 2024. Granting Non-AI Experts Creative Control Over AI Systems. In *The 37th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct '24)*, October 13–16, 2024, Pittsburgh, PA, USA. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3672539.3686714>

1 INTRODUCTION

While AI developers are well-equipped to design tools for their own lives, they are unfortunately not best equipped to design tools for the multitude of lives that their AI systems impact—across countless professions, tasks, communities, and cultures. As a result, we have seen myriad ways in which AI can perpetuate harmful biases and embed problematic values. Content moderation models censor discussion of marginalized identities while failing to shield users from thinly-veiled hate [4, 11]. Social media feed ranking models oriented around user engagement amplify political polarization, posing risks to the institution of democracy [10, 12, 17]. The situation is only growing more dire with the rise of large language models, which only represent the voices of a small group of technological experts [14, 15].

We need experts from the *domains and communities where AI is deployed* to design the AI systems that impact them, but this task is prohibitively difficult for non-AI experts. AI development requires substantial technical knowledge and considerable effort [19, 20].

Rather than requiring non-AI experts to become AI experts themselves, or requiring continued reliance on AI experts to lossily translate their intents into technical implementations, what if we could *redesign AI development tools* to match the language with which non-AI experts express their ideas? My research develops *systems*

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

UIST Adjunct '24, October 13–16, 2024, Pittsburgh, PA, USA

© 2024 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0718-6/24/10

<https://doi.org/10.1145/3672539.3686714>

for users to take creative control over AI by defining AI behavior in their own words, in terms of concepts they understand. For example, my work has allowed users to build a content moderation model by specifying concepts like “profanity” and “misogyny” that the model should consider, or to create a new feed ranking model by describing the kinds of divisive political content that should be downranked.

Advances in large language models allow us to communicate a much broader set of tasks and concepts with natural language, but LLMs are unpredictable and challenging to steer. To achieve our vision of AI, we need to deliver a match between a *specific user* and the *particular vision of AI behavior* that they want to achieve. For example, consider a content moderator developing a moderation model for their community. While they can prompt an LLM off-the-shelf to perform content moderation, this model would fail in practice because it lacks an understanding of this *particular* community’s unique moderation needs [3]. Who are the members of the community? What kind of language do they use? What does this moderator believe constitutes profanity, hate speech, or other potential harms? We can’t address these questions unless we redesign how we gather specific end users’ needs and how we instantiate these needs in models.

To tackle this challenge, our insight is that instead of building monolithic models, we need to support end users in creating many *narrow* models, which allow for greater control and interpretability. With narrow models, we have the capacity to learn specific, user-defined concepts rather than generic versions of concepts. We demonstrate how we can develop these narrow, user-specific models with a low amount of user effort in a variety of settings: by translating user expertise in written codebooks to instantiate novel AI objective functions; collecting a small set of user labels to support targeted AI auditing; and mining a user’s existing writing logs to develop customized LLM adapters for their own AI writing assistant.

When non-AI experts can design AI from start to finish—from objectives to application to evaluation—they can achieve outcomes not possible with systems owned by central AI experts. For example, by expressing their domain knowledge to design AI *objectives*, our Societal Objective Functions work authored new social media ranking algorithms based on political science constructs and successfully mitigated partisan animosity [5]. Our work on End-User Audits empowered everyday users to lead their own AI *evaluations*, which allowed them to surface underreported issues with Perspective API, a widely-used content moderation model [6]. Going further, the ongoing Brushes project grants ownership over the *entire process* from modeling to use to evaluation and repair. This approach allows end users to powerfully control the behavior of an LLM to faithfully mirror facets of their personal writing style, far beyond what base models could achieve. By changing the language with which we carry out AI development, we can help AI experts and non-experts alike to creatively control the entire AI lifecycle.

2 DESIGNING AI OBJECTIVE FUNCTIONS AROUND SOCIETAL VALUES

Concept-based abstractions allow non-AI experts to expressively translate their domain knowledge into AI *objectives*.

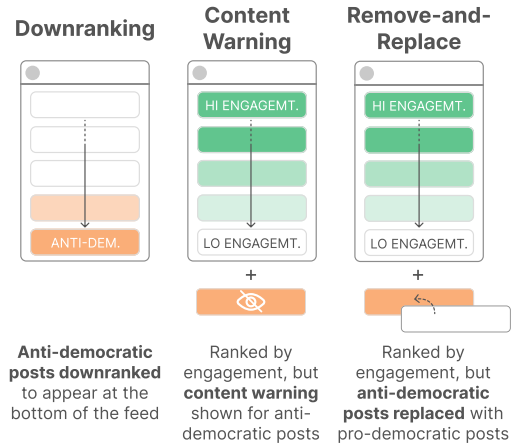


Figure 2: Societal objective functions can re-rank feeds based on political science concepts to mitigate partisan animosity.

Societal Objective Functions: Translating social science expertise. Many of the challenges of AI development are concerned not only with technical details, but with human behavior. For example, mounting evidence indicates that the AI systems that rank our social media feeds bear nontrivial responsibility for amplifying partisan animosity: negative thoughts, feelings, and behaviors toward political out-groups [10, 12, 17]. How ought we to design these AIs to mitigate partisan animosity as part of their objective functions? AI design problems like these are sociotechnical challenges that require social science expertise to design effective interventions.

To allow social scientists to author their own such interventions, we introduced a method for translating established, vetted social scientific constructs into AI objective functions, which we term *societal objective functions* [5]. We demonstrated the method with application to a political science construct related to anti-democratic attitudes. Traditionally, we have lacked observable outcomes to use to train such models — however, the social sciences have developed survey instruments and qualitative codebooks for these constructs, and their precision facilitates translation into detailed prompts for large language models.

By applying the societal objective functions method to encode political scientists’ domain knowledge on anti-democratic attitudes, we successfully created new social media feeds that significantly reduced users’ partisan animosity (Figure 2). Across three studies, we investigated how feeds reranked with alternative societal objective functions impacted partisan animosity and traditional platform engagement metrics. Starting first with a manual intervention ($N = 1,380$) and then replicating the same results with our AI intervention ($N = 558$), we found that feeds re-ranked with our societal objective function significantly reduced partisan animosity compared to a traditional engagement-based feed ($d = .25, p = .02$). These results held for both conservatives and liberals, and promisingly, did not compromise participants’ engagement level or ratings of their platform experience.

Societal objective functions present a novel strategy to draw on social science theory and methods to mitigate societal harms in social media AIs and AI systems more broadly. While we demonstrated the viability of this method with a specific construct of

democratic attitudes for social media AIs, our method points toward an approach for incorporating a much broader set of potential societal values into AI. Notably, today’s AI systems *already* encode values, but those values are typically implicit and defined by platforms and the technical experts who build them. Our work makes these values more explicit and tuneable such that non-AI experts can contribute their domain expertise to redefine the values of AI.

Model Sketching: Promoting model design space exploration. Grounding AI development in user-understandable concepts not only lends voice to non-AI experts, but also help AI experts to better reason about societal impacts. Machine learning practitioners often end up fixating on low-level technical decisions. When designing a content moderation model, it is easy to become caught up in questions like, “How would a transformer model perform if we train on these examples and use these hyperparameters?” Could we instead stay focused on questions like, “What factors—such as violence, racism, sexism, and profanity—should our model pay attention to when determining what content is hateful?” Could our AI development tools center our efforts not on how to *execute a solution*, but on *what kinds of solutions to pursue* in the first place?

Inspired by the practice of sketching in design, which distills ideas to their minimal representation, we introduced *model sketching*: a technical framework for iteratively and rapidly authoring functional approximations of a machine learning model’s decision-making logic [7]. We centered *concepts*, human-understandable factors that are relevant to a decision-making task, as the minimal representation for ML sketches. Leveraging advances in zero-shot learning with foundation models like GPT and CLIP, users can interactively author classifiers for arbitrary concepts using natural language. These concept classifiers can be incorporated into lightweight “sketch models” trained to perform tasks like classification and ranking. We instantiated this approach in the ModelSketchBook¹ system, which allows users to author ML models purely by specifying high-level text or image concepts that the model is expected to reason over (e.g., profanity, racism, politics, or violence in a multimodal content moderation task).

In an evaluation with 17 ML practitioners, model sketching reframed thinking from implementation to higher-level exploration. Starting from scratch with their own tools, AI experts initially focused on technical decisions like model architectures and performance metrics. With ModelSketchBook, they rapidly explored a diverse set of model designs (spanning 136 unique concepts ranging from “racism” and “Islamophobia” to “sarcasm” and “controversial”), and they brainstormed concepts based on potential user harms. This rapid iteration allowed participants to notice issues of dataset representativity, class imbalance, inconsistent labels—all in under half an hour, a fraction of the time ordinarily required to build a model. Changing the core abstractions of model development with ModelSketchBook helps AI experts to proactively consider the human impact and ethical implications of their models.

¹ModelSketchBook is available as an open-source Python package: <https://github.com/StanfordHCL/ModelSketchBook>

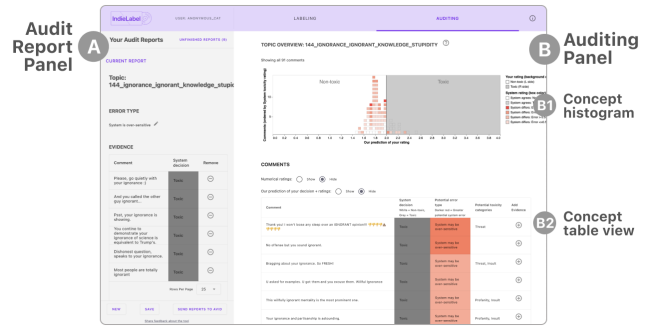


Figure 3: IndieLabel aids end users in leading audits by surfacing concepts where they likely disagree with the model.

3 EMPOWERING END USERS TO LEAD EVALUATIONS OF HARMFUL AI BEHAVIOR

User-defined concepts are also a powerful scaffold for non-AI experts to *evaluate* AI. For example, algorithm audits are a critical method to hold algorithmic systems accountable, and audits have surfaced critical issues in domains like healthcare, housing, and employment [1, 2, 13]. However, because algorithm audits are led by a small group of technical experts, they are necessarily limited to the hypotheses that experts think to test. End-users could help to expand the purview of audits, as they hold rich knowledge of their communities and the ways in which algorithms impact them. However, it is challenging for end users to lead their own audits, which require substantial effort to label thousands of system outputs.

To achieve the goal of *end-user algorithm audits*, system-scale audits led by individual, non-technical users, we introduced IndieLabel,² a tool that substantially lowers the barrier to entry [6]. From a handful of user labels, our approach automatically identifies candidate concepts for which a user most likely disagrees with the current model, allowing users to lead audits in a matter of minutes. We accomplish this with a collaborative filtering technique that leverages the model’s own disaggregated training data to project from a small number of end user labels onto the full test set. With these estimated user labels in hand, we can identify concepts (e.g., “feminism,” “religion,” “scam”) where the user’s opinions most strongly diverge from that of the model. We direct the user’s attention to just these areas where the model is performing worst for them and where their perspective is most needed. With as few as 20 labels from a user, the system can estimate their ratings on over 100,000 examples and scaffold them in hypothesis generation, evidence identification, and results communication (Figure 3).

In an evaluation of IndieLabel with 17 non-technical participants, end-user auditors successfully led audits of Perspective API, a widely-used toxicity model, within a half-hour auditing period. Participants both replicated issues that formal, expert-led audits had previously identified and also raised previously underreported issues, such as under-flagging for veiled forms of hate and over-flagging for slurs reclaimed by marginalized communities. IndieLabel was publicly deployed by the AI Risk and Vulnerability Alliance, a civil society organization focused on empowering communities to surface AI vulnerabilities.

²IndieLabel is open-source: <https://github.com/StanfordHCL/indie-label>

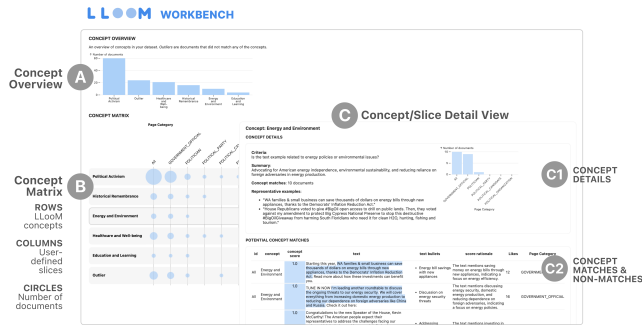


Figure 4: LLoOM induces high-level, interpretable concepts from unstructured text, aiding theory-driven data analysis.

My work has found that non-AI experts can play a critical role in expanding the scope of AI evaluation, whether they contribute their individual perspectives actively or passively. Our Sociotechnical Auditing system allows users to passively contribute to AI audits by embedding audits in the web browser and coordinating longitudinal interventions on hundreds of users’ ordinary web browsing [8]. Applying this approach to audit ad targeting models, we found that their efficacy may stem from repeated exposure rather than inherent user alignment. Meanwhile, our LLoOM³ concept induction system allows non-AI experts themselves to steer data analysis and automatically extract interpretable concepts from unstructured text [9]. With our system, it is possible to conduct theory-driven analyses of a wide range of models, such as content moderation models, social media feed ranking models, and large language models (Figure 4). LLoOM allowed data analysts to discover new insights even from familiar datasets. Non-AI experts can see around corners and fill in major gaps in AI evaluation, especially when they are granted a high level of control and creative freedom to explore the questions they find most important.

4 INDUCING END-USER CONTROLS FOR LLM GENERATION

Finally, concepts can be a powerful lever for non-AI experts themselves to fully define AI behavior from *end-to-end*. As large language models become ever more popular, worrisome patterns are emerging: what happens when models produce fluent but bland writing, or when they produce arguments that sound plausible but don’t align with our views? And what happens when everyone in the world is assisted by the same models, which tend towards the same styles, arguments, and ideas? We’re trending towards a future of AI that allows users to efficiently carry out more tasks than ever—but only in the same way as everyone else.

What if instead our guiding metaphor of AI interaction was not monolithic, but pluralistic? How could powerful modern AI systems not lead us to efficiently converge, but to creatively diverge—to maintain and even amplify what is uniquely “you”? Our vision is to induce a large number of highly specific, customized model adapters tailored to the end user, which we term *brushes*. Building a single monolithic model for each user yields little control: how does a user convey that they want the model to behave one way

in one instance, but another way the next time the task arises? We instead learn a *space* of brushes for each user that they can control and apply in response to any need.

State-of-the-art language models hold a plethora of possible behaviors, but the challenge lies in finding and activating the pocket of relevant behavior from a massive model. We can already mine salient facets of an end user’s behavior from their existing interaction logs and writings. Our insight is that once we have a handful of these demonstrations, we have enough useful signal to teach a language model to isolate and reproduce this desired behavior. In particular, we demonstrate two methods to instantiate brushes, neither of which require additional user effort: a lightweight representation steering method (learning an inference-time intervention on a model’s internal representation) [18] and a heavier weight low-rank adapter (LoRA) method (learning modified weights on top of a frozen base model) [16]. With a pool of these fine-tuned LLM variants, end users can access facets of model behavior that are relevant to their demonstrated needs.

We are building the Brushes system to instantiate this vision, focusing on writing assistance as a case study. The system observes the writing of an end user to learn a set of personalized LLM brushes—each representing an interpretable concept (e.g., “Friendly Informal Tone” or “Professional Networking”) defined by the user’s own behavior. Just as a standard language model can carry out instructions (e.g., “write an email to schedule a meeting with my colleague,” “rewrite this CHI abstract”), our system can automatically orchestrate a relevant set of brushes under the hood to carry out tasks for a user. However, in contrast to a black-box, monolithic LLM, since our system carries out writing tasks by composing a user’s personalized brushes (e.g., their own brand of “Academic Voice,” or their typical style in “CHI Submissions”), the user can look under the hood to inspect, swap, and repair brushes to their liking. With these precise levers into LLM behavior, our system is equipped to produce many potential versions of the user, and the user has control to summon and deploy these facets of themselves as they wish.

5 CONCLUSION

My research demonstrates that non-AI experts can fully own and define AI behavior from end-to-end. Moving beyond a single, general-purpose, third-party model and towards many narrow, user-defined models makes it possible for users to understand and creatively control AI. By expressly designing AI systems around user-understandable concepts, AI experts and non-experts alike can think, speak, and act to address the far-reaching impacts of AI—for themselves, their communities, and society at large.

ACKNOWLEDGMENTS

I am incredibly grateful to my advisors, Michael Bernstein and James Landay, for supporting me throughout the course of the Ph.D. and helping me to develop my own voice in research. Thank you to my many mentors, collaborators, labmates, and friends who have shared their wisdom and time to support my work.

REFERENCES

- [1] Joshua Asplund, Motahare Eslami, Hari Sundaram, Christian Sandvig, and Karrie Karahalios. 2020. Auditing race and gender discrimination in online housing

³LLoOM is available as an open-source Python package: <https://github.com/michelle123lam/lloom>

- markets. In *Proceedings of the International AAAI Conference on Web and Social Media*, Vol. 14. 24–35.
- [2] Jack Bandy. 2021. Problematic Machine Behavior: A Systematic Literature Review of Algorithm Audits. *Proc. ACM Hum.-Comput. Interact.* 5, CSCW1, Article 74 (apr 2021), 34 pages. <https://doi.org/10.1145/3449148>
 - [3] Robert Gorwa, Reuben Binns, and Christian Katzenbach. 2020. Algorithmic content moderation: Technical and political challenges in the automation of platform governance. *Big Data & Society* 7, 1 (2020), 2053951719897945. <https://doi.org/10.1177/2053951719897945> arXiv:<https://doi.org/10.1177/2053951719897945>
 - [4] Oliver L. Haimson, Daniel Delmonaco, Peipei Nie, and Andrea Wegner. 2021. Disproportionate Removals and Differing Content Moderation Experiences for Conservative, Transgender, and Black Social Media Users: Marginalization and Moderation Gray Areas. *Proc. ACM Hum.-Comput. Interact.* 5, CSCW2, Article 466 (Oct 2021), 35 pages. <https://doi.org/10.1145/3479610>
 - [5] Chenyan Jia*, Michelle S. Lam*, Minh Chau Mai, Jeffrey T. Hancock, and Michael S. Bernstein. 2024. Embedding Democratic Values into Social Media AIs via Societal Objective Functions. *Proc. ACM Hum.-Comput. Interact.* 8, CSCW1, Article 163 (Apr 2024), 36 pages. <https://doi.org/10.1145/3641002>
 - [6] Michelle S. Lam, Mitchell L. Gordon, Dana* Metaxa, Jeffrey T. Hancock, James A. Landay, and Michael S. Bernstein. 2022. End-User Audits: A System Empowering Communities to Lead Large-Scale Investigations of Harmful Algorithmic Behavior. *Proc. ACM Hum.-Comput. Interact.* 6, CSCW2, Article 512 (Nov 2022), 34 pages. <https://doi.org/10.1145/3555625>
 - [7] Michelle S. Lam, Zixian Ma, Anne Li, Izequiel Freitas, Dakuo Wang, James A. Landay, and Michael S. Bernstein. 2023. Model Sketching: Centering Concepts in Early-Stage Machine Learning Model Design. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 741, 24 pages. <https://doi.org/10.1145/3544548.3581290>
 - [8] Michelle S. Lam, Ayush Pandit, Colin H. Kalicki, Rachit Gupta, Poonam Sahoo, and Dana* Metaxa. 2023. Sociotechnical Audits: Broadening the Algorithm Auditing Lens to Investigate Targeted Advertising. *Proc. ACM Hum.-Comput. Interact.* 7, CSCW2, Article 360 (Oct 2023), 37 pages. <https://doi.org/10.1145/3610209>
 - [9] Michelle S. Lam, Janice Teoh, James A. Landay, Jeffrey Heer, and Michael S. Bernstein. 2024. Concept Induction: Analyzing Unstructured Text with High-Level Concepts Using LLoM. In *Proceedings of the CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 766, 28 pages. <https://doi.org/10.1145/3613904.3642830>
 - [10] Philipp Lorenz-Spreen, Lisa Oswald, Stephan Lewandowsky, and Ralph Hertwig. 2023. A systematic review of worldwide causal and correlational evidence on digital media and democracy. *Nature human behaviour* 7, 1 (2023), 74–101.
 - [11] Brandeis Marshall. 2021. *Algorithmic Misogynoir in Content Moderation Practice*. Technical Report. Technical Report. Heinrich-Böll-Stiftung.
 - [12] Smitha Milli, Micah Carroll, Sashrika Pandey, Yike Wang, and Anca D Dragan. 2023. Twitter's Algorithm: Amplifying Anger, Animosity, and Affective Polarization. *arXiv preprint arXiv:2305.16941* (2023).
 - [13] Ziad Obermeyer, Brian Powers, Christine Vogeli, and Sendhil Mullainathan. 2019. Dissecting racial bias in an algorithm used to manage the health of populations. *Science* 366, 6464 (2019), 447–453. <https://doi.org/10.1126/science.aax2342> arXiv:<https://www.science.org/doi/pdf/10.1126/science.aax2342>
 - [14] Michael J. Ryan, William Held, and Diyi Yang. 2024. Unintended Impacts of LLM Alignment on Global Representation. arXiv:2402.15018 [cs.CL]
 - [15] Shibani Santurkar, Esin Durmus, Faisal Ladhak, Cinoo Lee, Percy Liang, and Tatsunori Hashimoto. 2023. Whose opinions do language models reflect?. In *Proceedings of the 40th International Conference on Machine Learning* (Honolulu, Hawaii, USA) (ICML '23). JMLR.org, Article 1244, 34 pages.
 - [16] Omar Shaikh, Michelle Lam, Joey Hejna, Yijia Shao, Michael Bernstein, and Diyi Yang. 2024. Show, Don't Tell: Aligning Language Models with Demonstrated Feedback. arXiv:2406.00888 [cs.CL]
 - [17] Petter Törnberg. 2022. How digital media drive affective polarization through partisan sorting. *Proceedings of the National Academy of Sciences* 119, 42 (2022), e2207159119.
 - [18] Zhengxuan Wu, Aryaman Arora, Zheng Wang, Atticus Geiger, Dan Jurafsky, Christopher D. Manning, and Christopher Potts. 2024. ReFT: Representation Finetuning for Language Models. arxiv.org/abs/2404.03592
 - [19] Qian Yang, Aaron Steinfeld, Carolyn Rosé, and John Zimmerman. 2020. Re-examining Whether, Why, and How Human-AI Interaction Is Uniquely Difficult to Design. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3313831.3376301>
 - [20] Qian Yang, Jina Suh, Nan-Chen Chen, and Gonzalo Ramos. 2018. Grounding Interactive Machine Learning Tool Design in How Non-Experts Actually Build Models. In *Proceedings of the 2018 Designing Interactive Systems Conference* (Hong Kong, China) (DIS '18). Association for Computing Machinery, New York, NY, USA, 573–584. <https://doi.org/10.1145/3196709.3196729>